John C. Flournoy, PhD

Research Scientist, Stats Maven

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Education

Ph.D., Psychology, University of Oregon
Dissertation: *Adolescent Social Motives: Measurement and Implications* 2018

M.S., Psychology, University of Oregon 2014

B.A., Cognitive Science, University of California, Berkeley 2005

Work Experience

Research and Data Science Consultant, Independent 2018–present

- Clients have included Meru Health (digital mental health analyses), Harvard (HCPD neuroimaging), and academic teams seeking psychometric and causal inference expertise

Principal Research Scientist, Developer Success Lab, Pluralsight 2024–2025

- Lead scientist on analysis of yearly ticket data from over 11k developers
- Drove organizational insights across multiple research projects
- Spearheaded community engagement overlay journal at dsl.pubpub.org

Research Associate, Harvard University, Cambridge, MA 2018–2024

- Lead analyst on intensive longitudinal fMRI neuroimaging study of mechanisms linking stress to psychopathology in adolescents
- Lead author of report on fMRI measurement reliability
- Supported co-authors on intensive longitudinal digital phenotyping analysis of stress, sleep, digital communication, and physical activity
- Lead analyst of task-based fMRI for multi-site adolescent cognitive control study (NIMH U01-MH109589, $17.1M)
- Developed digital phenotyping data processing for NIMH R37-MH119194 ($9M)
- Provided methodology and scientific computing support across two lab groups
- note: post-doctoral scholar from 2018-2020

Graduate Research Fellow, University of Oregon, Eugene, OR 2012–2018

- Developed hierarchical Bayesian model of reinforcement learning for adolescent social motives
- Collected data from 300+ participants, including foster-care-involved adolescents
- Dissertation: *Adolescent Social Motives: Measurement and Implications*
- Developed expertise in longitudinal personality, fMRI, and cognitive task data using multilevel and structural equation modeling
- Validated new self-report and task measures, and evaluated validity and reliability of existing measures

Research Coordinator, Stanford University, Stanford, CA 2009–2012

- Coordinated Simons Foundation-funded study on sleep problems in autism using ambulatory polysomnography
- Site coordinator for clinical trial testing a novel PET marker for β-amyloid in dementia patients

Publications

Cheng, T. W., Mair, P., Curtis, M. T., Baum, G. L., Flournoy, J. C., Wang, S., Glasser, M. F., Barch, D. M., & Somerville, L. H. (2026). Longitudinal changes in T1w/T2w estimates of cortical myelin with age and pubertal timing. *Cerebral Cortex*, *36*(2), bhag004. https://doi.org/10.1093/cercor/bhag004

Curtis, M., Yendiki, A., Cheng, T., Omary, A., Flournoy, J. C., Kandala, S., Sanders, A. F., Harms, M. P., Somerville, L. H., & Barch, D. M. (2026). Disentangling the unique associations of age, pubertal stage, and pubertal hormones with white matter microstructure in childhood and adolescence. *Imaging Neuroscience*, *4*, IMAG.a.1270. https://doi.org/10.1162/IMAG.a.1270

Chen, Y. Y., Rosenbaum, G. M., Fan, H., Flournoy, J. C., Li, T., Cegarra, L., Youssoufian, D. A., Grad-Freilich, M. J., Kordyban, L. E., Mair, P., & Somerville, L. H. (2025). Social Contexts Requiring Adjudication between Self- and Peer-Interest Differentially Alter Risk Preferences Across Adolescence. *Open Mind*, *9*, 540–558. https://doi.org/10.1162/opmi_a_00201

Flournoy, J. C., Lee, C. S., Wu, M., & Hicks, C. M. (2025). No silver bullets: Why understanding software cycle time is messy, not magic. *Empirical Software Engineering*, *30*(6), 174. https://doi.org/10.1007/s10664-025-10735-w

Guazzelli Williamson, V., Barendse, M. E. A., Chavez, S. J., Flournoy, J. C., Cheng, T. W., Cosme, D., Byrne, M. L., Allen, N. B., & Pfeifer, J. H. (2025). A longitudinal neuroimaging study of adolescent girls' mentalizing and perspective-taking tendencies. *Developmental Cognitive Neuroscience*, *72*, 101526. https://doi.org/10.1016/j.dcn.2025.101526

Rakesh, D., Flournoy, J. C., & McLaughlin, K. A. (2025). Associations between socioeconomic status and mental health trajectories during early adolescence: Findings from the Adolescent Brain Cognitive Development study. *JCPP Advances*, *5*(4), e70001. https://doi.org/10.1002/jcv2.70001

Chen, Y. Y., Rosenbaum, G. M., Fan, H., Li, T., Flournoy, J., Mair, P., & Somerville, L. (2024). How beliefs around peers' risk preferences get incorporated into adolescents' decision making. *Proceedings of the Annual Meeting of the Cognitive Science Society*, *46*(0). Retrieved January 28, 2025, from https://escholarship.org/uc/item/6c42379z

Curtis, M., Flournoy, J. C., Kandala, S., Sanders, A. F. P., Harms, M. P., Omary, A., Somerville, L. H., & Barch, D. M. (2024). Disentangling the unique contributions of age, pubertal stage, and pubertal hormones to brain structure in childhood and adolescence. *Developmental Cognitive Neuroscience*, *70*, 101473. https://doi.org/10.1016/j.dcn.2024.101473

Flournoy, J. C., Bryce, N. V., Dennison, M. J., Rodman, A. M., McNeilly, E. A., Lurie, L. A., Bitran, D., Reid-Russell, A., Vidal Bustamante, C. M., Madhyastha, T., & McLaughlin, K. A. (2024). A precision neuroscience approach to estimating reliability of neural responses during emotion processing: Implications for task-fMRI. *NeuroImage*, *285*, 120503. https://doi.org/10.1016/j.neuroimage.2023.120503

Heffer, T., Flournoy, J. C., Baum, G. L., & Somerville, L. H. (2024). Examining the Association between Punishment and Reward Sensitivity and Response Inhibition to Previously-Incentivized Cues across Development. *Journal of Youth and Adolescence*, *53*(6), 1341–1354. https://doi.org/10.1007/s10964-024-01966-z

Rith-Najarian, L. R., Gong-Guy, E., Flournoy, J. C., & Chavira, D. A. (2024). Randomized controlled trial of a web-based program for preventing anxiety and depression in university students. *Journal of Consulting and Clinical Psychology*, *92*(1), 1–15. https://doi.org/10.1037/ccp0000843

Xu, B., Dall'Aglio, L., Flournoy, J., Bortsova, G., Tervo-Clemmens, B., Collins, P., de Bruijne, M., Luciana, M., Marquand, A., Wang, H., Tiemeier, H., & Muetzel, R. L. (2024). Limited generalizability of multivariate brain-based dimensions of child psychiatric symptoms. *Communications Psychology*, *2*(1), 1–14. https://doi.org/10.1038/s44271-024-00063-y

Byrne, M. L., Vijayakumar, N., Chavez, S. J., Flournoy, J. C., Cheng, T. W., Mills, K. L., Barendse, M. E. A., Mobasser, A., Flannery, J. E., Nelson, B. W., Wang, W., Shirtcliff, E. A., Allen, N. B., & Pfeifer, J. H. (2023). Associations between multi-method latent factors of puberty and brain structure in adolescent girls. *Developmental Cognitive Neuroscience*, *60*, 101228. https://doi.org/10.1016/j.dcn.2023.101228

Grisanzio, K. A., Flournoy, J. C., Mair, P., & Somerville, L. H. (2023). Shifting qualities of negative affective experience through adolescence: Age-related change and associations with functional outcomes. *Emotion*, *23*(1), 278–288. https://doi.org/10.1037/emo0001079

McCormick, E. M., Byrne, M. L., Flournoy, J. C., Mills, K. L., & Pfeifer, J. H. (2023). The Hitchhiker's guide to longitudinal models: A primer on model selection for repeated-measures methods. *Developmental Cognitive Neuroscience*, *63*, 101281. https://doi.org/10.1016/j.dcn.2023.101281

McLaughlin, K. A., Weissman, D. G., & Flournoy, J. (2023). Challenges with Latent Variable Approaches to Operationalizing Dimensions of Childhood adversity – a Commentary on Sisitsky et al. (2023). *Research on Child and Adolescent Psychopathology*, *51*(12), 1809–1811. https://doi.org/10.1007/s10802-023-01114-4

Barendse, M. E. A., Byrne, M. L., Flournoy, J. C., McNeilly, E. A., Guazzelli Williamson, V., Barrett, A.-M. Y., Chavez, S. J., Shirtcliff, E. A., Allen, N. B., & Pfeifer, J. H. (2022). Multimethod assessment of pubertal timing and associations with internalizing psychopathology in early adolescent girls. *Journal of Psychopathology and Clinical Science*, *131*(1), 14–25. https://doi.org/10.1037/abn0000721

Baum, G. L., Flournoy, J. C., Glasser, M. F., Harms, M. P., Mair, P., Sanders, A. F. P., Barch, D. M., Buckner, R. L., Bookheimer, S., Dapretto, M., Smith, S., Thomas, K. M., Yacoub, E., Essen, D. C. V., & Somerville, L. H. (2022). Graded Variation in T1w/T2w Ratio during Adolescence: Measurement, Caveats, and Implications for Development of Cortical Myelin. *Journal of Neuroscience*, *42*(29), 5681–5694. https://doi.org/10.1523/JNEUROSCI.2380-21.2022

Byrne, M., Allen, N. B., Barendse, M. E. A., Barrett, A.-M., Chavez, S. J., Cheng, T., Flournoy, J. C., McNeilly, E. A., Shirtcliff, E. A., Vijayakumar, N., Williamson, V. G., & Pfeifer, J. (2022). Pubertal Hormones as a Measurement Method of Pubertal Stage and Timing Related to Internalizing Psychopathology in Adolescent Girls. *Biological Psychiatry*, *91*(9), S64–S65. https://doi.org/10.1016/j.biopsych.2022.02.180

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Bryce, N. V., Flournoy, J. C., Guassi Moreira, J. F., Rosen, M. L., Sambook, K. A., Mair, P., & McLaughlin, K. A. (2021). Brain parcellation selection: An overlooked decision point with meaningful effects on individual differences in resting-state functional connectivity. *NeuroImage*, *243*, 118487. https://doi.org/10.1016/j.neuroimage.2021.118487

Flournoy, J. (2021, September 16). *A comment on Ellwood-Lowe et al. (2020/2021): Presentation and discussion of results should take the lead from the preregistration*. https://doi.org/10.31234/osf.io/23qsw

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Hughes, B. T., Flournoy, J. C., & Srivastava, S. (2021). Is perceived similarity more than assumed similarity? An interpersonal path to seeing similarity between self and others. *Journal of Personality and Social Psychology*, *121*(1), 184–200. https://doi.org/10.1037/pspp0000369

López-Vicente, M., Agcaoglu, O., Pérez-Crespo, L., Estévez-López, F., Heredia-Genestar, J. M., Mulder, R. H., Flournoy, J. C., van Duijvenvoorde, A. C. K., Güroğlu, B., White, T., Calhoun, V., Tiemeier, H., & Muetzel, R. L. (2021). Developmental Changes in Dynamic Functional Connectivity From Childhood Into Adolescence. *Frontiers in Systems Neuroscience*, *15*. https://doi.org/10.3389/fnsys.2021.724805

Nook, E. C., Flournoy, J. C., Rodman, A. M., Mair, P., & McLaughlin, K. A. (2021). High Emotion Differentiation Buffers Against Internalizing Symptoms Following Exposure to Stressful Life Events in Adolescence: An Intensive Longitudinal Study. *Clinical Psychological Science*, 2167702620979786. https://doi.org/10.1177/2167702620979786

Rodman, A. M., Vidal Bustamante, C. M., Dennison, M. J., Flournoy, J. C., Coppersmith, D. D. L., Nook, E. C., Worthington, S., Mair, P., & McLaughlin, K. A. (2021). A Year in the Social Life of a Teenager: Within-Persons Fluctuations in Stress, Phone Communication, and Anxiety and Depression. *Clinical Psychological Science*, 216770262199180 4. https://doi.org/10.1177/2167702621991804

Andrews, J. L., Mills, K. L., Flournoy, J. C., Flannery, J. E., Mobasser, A., Ross, G., Durnin, M., Peake, S., Fisher, P. A., & Pfeifer, J. H. (2020). Expectations of Social Consequences Impact Anticipated Involvement in Health-Risk Behavior During Adolescence. *Journal of Research on Adolescence: The Official Journal of the Society for Research on Adolescence*, *30*(4), 1008–1024. https://doi.org/10.1111/jora.12576

Barendse, M. E. A., Cosme, D., Flournoy, J. C., Vijayakumar, N., Cheng, T. W., Allen, N. B., & Pfeifer, J. H. (2020). Neural correlates of self-evaluation in relation to age and pubertal development in early adolescent girls. *Developmental Cognitive Neuroscience*, *44*, 100799. https://doi.org/10.1016/j.dcn.2020.100799

Barendse, M. E. A., Vijayakumar, N., Byrne, M. L., Flannery, J. E., Cheng, T. W., Flournoy, J. C., Nelson, B. W., Cosme, D., Mobasser, A., Chavez, S. J., Hval, L., Brady, B., Nadel, H., Helzer, A., Shirtcliff, E. A., Allen, N. B., & Pfeifer, J. H. (2020). Study Protocol: Transitions in Adolescent Girls (TAG). *Frontiers in Psychiatry*, *10*. https://doi.org/10.3389/fpsyt.2019.01018

Cheng, T. W., Vijayakumar, N., Flournoy, J. C., Op de Macks, Z., Peake, S. J., Flannery, J. E., Mobasser, A., Alberti, S. L., Fisher, P. A., & Pfeifer, J. H. (2020). Feeling left out or just surprised? Neural correlates of social exclusion and overinclusion in adolescence. *Cognitive, Affective, & Behavioral Neuroscience*, *20*(2), 340–355. https://doi.org/10.3758/s13415-020-00772-x

Flournoy, J. C., Vijayakumar, N., Cheng, T. W., Cosme, D., Flannery, J. E., & Pfeifer, J. H. (2020). Improving practices and inferences in developmental cognitive neuroscience. *Developmental Cognitive Neuroscience*, *45*, 100807. https://doi.org/10.1016/j.dcn.2020.100807

Vidal Bustamante, C. M., Rodman, A. M., Dennison, M. J., Flournoy, J. C., Mair, P., & McLaughlin, K. A. (2020). Within-person fluctuations in stressful life events, sleep, and anxiety and depression symptoms during adolescence: A multiwave prospective study. *Journal of Child Psychology and Psychiatry*, *61*(10), 1116–1125. https://doi.org/10.1111/jcpp.13234

Vijayakumar, N., Flournoy, J. C., Mills, K. L., Cheng, T. W., Mobasser, A., Flannery, J. E., Allen, N. B., & Pfeifer, J. H. (2020). Getting to know me better: An fMRI study of intimate and superficial self-disclosure to friends during adolescence. *Journal of Personality and Social Psychology*, *118*(5), 885–899. https://doi.org/10.1037/psp a0000182

Byrne, M. L., Chavez, S., Vijayakumar, N., Cheng, T. W., Flournoy, J., Barendse, M., Shirtcliff, E. A., Allen, N., & Pfeifer, J. (2019). Multi-method confirmatory factor analyses of puberty in early adolescent girls. https://doi.org/10.31234/osf.io/pue6f

Flournoy, J. (2019, January 11). *Adolescent Social Motives: Measurement and Implications*. Retrieved January 6, 2020, from https://scholarsbank.uoregon.edu/xmlui/handle/1794/24216

Ludwig, R. M., Flournoy, J. C., & Berkman, E. T. (2019). Inequality in personality and temporal discounting across socioeconomic status? Assessing the evidence. *Journal of Research in Personality*, *81*, 79–87. https://doi.org/10.1016/j.jrp.2019.05.003

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Thalmayer, A. G., Saucier, G., Srivastava, S., Flournoy, J. C., & Costello, C. K. (2019). Ethics-relevant values in adulthood: Longitudinal findings from the life and time study. *Journal of Personality*, *87*(6), 1119–1135. https://doi.org/10.1111/jopy.12462

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